

SynapseLab

An Interactive Simulation of Memory, Synaptic Plasticity and Interference

DataForge 2026 — Project Blog

1. Project Concept and Problem

SynapseLab is an interactive web-based educational simulation that visualizes how a simplified network can learn patterns, strengthen associations, and become less reliable when previously learned information is disrupted by interference. The project addresses a difficult-to-visualize question: how can changes in connections between artificial neurons be used to demonstrate the basic intuition behind learning and memory? Instead of presenting memory as a black box, SynapseLab exposes the changing connection strengths so users can observe the process.

2. Neuroscience Motivation

The project is inspired by the neuroscience concept of synaptic plasticity—the ability of synapses to change their strength as a consequence of activity and experience. Research links activity-dependent changes in synaptic transmission with learning and memory. For example, Huang et al. (2023) reported that neuronal SNAP-23 is important for long-term potentiation and spatial memory in experimental models. Other research describes replay and offline synaptic plasticity as mechanisms associated with memory consolidation. SynapseLab does not reproduce the biological brain; it uses these ideas as conceptual inspiration for a transparent computational model.

3. How the Interactive Experiment Works

The user interacts with a small artificial memory system through the website. A pattern is selected and presented to the model. During learning, the system adjusts internal connection weights so that the selected pattern becomes associated with a particular response. During recall, the system compares the current input with the learned representations and produces a predicted pattern. Users can then introduce interference and observe how recall changes. This makes the relationship between learning, stored weights and retrieval visible rather than hidden inside a large machine-learning model.

4. CAT / DOG / BIRD / CAR Patterns

SynapseLab uses four simple symbolic classes: CAT, DOG, BIRD and CAR. They provide intuitive labels for four distinct learned patterns. The labels are not intended to represent a real-world image classifier or a clinically meaningful memory test. They are controlled experimental categories that make it easy to compare learning and recall before and after interference.

5. The 40% Interference Experiment

A central experiment applies 40% interference to the learned representation. Conceptually, interference introduces competing or altered information into the same simplified memory space. The purpose is to demonstrate that when stored associations are perturbed, the model may become less certain or may recall a different pattern. The 40% value is a simulation parameter chosen to make the effect visible and repeatable; it is not a biological measurement and should not be interpreted as “40% memory loss” in humans.

6. Scientific Evidence vs. Simulation

SynapseLab clearly separates evidence from analogy. Scientific literature supports the broad motivation that synaptic plasticity is involved in learning and memory. The particular four-class experiment, weight-update behavior, and 40% interference setting are computational design choices made for this project. Therefore, the website should be understood as an educational simulation rather than a neuroscience experiment, medical diagnostic system, or validated cognitive model.

7. Technology Stack

The project is implemented as a modern web application using Next.js and TypeScript, with the model logic organized in reusable TypeScript modules. The interface is deployed online through Vercel, while the source code is maintained in a public GitHub repository. The architecture emphasizes transparency: users can interact with the experiment while

the underlying simplified model remains understandable and reproducible.

8. Innovation and Potential Impact

The main value of SynapseLab is interpretability and accessibility. Students can interact with a memory-like model and immediately see the effect of learning and interference instead of only reading about neural plasticity. The same framework could later be extended with more patterns, adjustable learning rates, multiple interference schedules, visualization of individual synaptic weights, noise controls, and quantitative recall curves. This could make SynapseLab useful as a classroom demonstration or a starting point for experiments in computational neuroscience and explainable AI.

9. Limitations

The model is intentionally simplified. Biological memory involves many brain regions, cell types, molecular pathways, temporal scales and feedback mechanisms that are not represented here. The four pattern labels are symbolic, and the interference percentage is a simulation control rather than a scientifically calibrated human-memory metric. Results from SynapseLab therefore cannot be generalized directly to human cognition or medical conditions. A stronger research version would require validated datasets, explicit model comparisons, controlled experiments, statistical evaluation and biological grounding.

10. Links

Live website: <https://synapse-lab-three.vercel.app>

GitHub repository: <https://github.com/sagarpandare21/SynapseLab>

The GitHub repository is the public source-code link; the live website is the demonstration link for judges.

Research References

Huang M. et al. (2023). *Neuronal SNAP-23 is critical for synaptic plasticity and spatial memory independently of NMDA receptor regulation*. iScience, 26(5), 106664. PMID: 37168570. DOI: 10.1016/j.isci.2023.106664.

Offline neuronal activity and synaptic plasticity during sleep and memory consolidation. (2023). Review of evidence connecting neural replay, synaptic plasticity and long-term memory. PMID: 36584924.

Spike timing-dependent plasticity and memory. (2023). Review discussing forms of synaptic plasticity and their proposed role in memory formation. PMID: 36924615.

The molecular memory code and synaptic plasticity: A synthesis. (2023). Computational and biological perspectives on synapses as components of memory storage and inference. PMID: 36610586.

Submission note: SynapseLab is presented as an educational computational simulation inspired by established neuroscience concepts. Scientific claims are intentionally separated from the project's own simulation assumptions and design parameters.

Prepared for DataForge 2026 | SynapseLab | Public project links included above